Modeling Traffic Accidents in Wooster, Ohio using XGBoost Machine Learning Analysis

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1 Introduction

Traffic management remains a crucial issue in large cities, especially rapidly developing cities without extensive urban planning. Poor traffic management results in congestion, unnecessary emissions, and accidents that impact the safety of the city's residents [5]. In order to manage the flow of traffic within a city, many cities turn to data-driven approaches that inform urban planning.

"Smart cities" refer to cities that utilize data and computational techniques to improve the functioning of a city. For example, a smart city may use location services on cell phones to determine the best times and places to collect waste within a city. Intelligent traffic systems, a crucial component of smart cities, aggregate and analyze data to manage traffic within a city. Intelligent traffic systems may make use of traffic data, such as data gathered from Uber or from sensors placed around the city, to model traffic in real time. These traffic models can then be used to create software that responds to current traffic flows and decreases congestion, such as software like Google Maps that re-routes traffic to less congested areas. Urban planners, public sector transportation employees, politicians, and everyday people rely on accurate traffic models to make informed decisions about new transit routes or traffic policy.

Traffic accident analysis is an important subsection of traffic management. Motor vehicle crashes are the second leading cause of death in the United States among people ages 1-44, with traffic fatalities causing about 44,000 deaths in 2022 [14]. Traffic accident modeling utilizes traffic accident data to determine when and where traffic accidents will occur. This data can be used by city planners to create policy geared towards decreasing traffic accidents [8]. Intelligent traffic systems additionally use traffic models to reroute traffic away from accident-prone areas.

This study aims to provide better insight into traffic accidents in small American towns, including factors that can influence the likelihood and severity of a traffic accident. Using XGBoost analysis, this study will analyze a dataset of roughly 10,000 traffic accidents in Wooster, Ohio from 2021-2026 to model the likelihood of accidents, the severity of the accident, and determine which factors most contributed to causing a traffic accident. The results from this study will provide insight into what factors cause traffic accidents, where these accidents, providing information to traffic officials on strategic decisions that can be made to enhance public safety.

2 Background

The decision tree machine learning models of XGBoost and Random Forest are frequently used to model traffic in the literature [3]. Decision trees are simple machine learning algorithms that split into smaller, more specific branches, and come to a decision by tracing a path down the tree. In order to figure out which path to go down, the decision tree can be thought of as answering a series of questions. As an example, a decision tree may have been created to decide whether or not a student should study. The first branch may ask, "What time is it?" The tree may branch into two separate paths: one for if the time is between 11 pm and 6 am, and one for if it is 6am-11pm. Then, it may ask another question, "Do you have an exam tomorrow?" Here, the tree will split off into four paths: One for if it is nighttime and there is an exam, one for if it is nighttime and there is not an exam, one for if it is daytime and there is an exam, and one for if it is daytime and there is not an exam tomorrow. The decision tree may say that you should study if it is daytime and you have an exam, but not under any other condition.

In order for the decision tree to decide where to branch (say, maybe the first question should split at 5 am and 12 pm rather than 6 am and 11 pm), the trees are trained on a large body of data. The difference between XGBoost and Random Forest lies in the way that each new tree decision tree is created. Random Forest follows the "bagging" technique, which means that each tree is trained individually on a different subset of the data. XGBoost, in contrast, follows the "boosting" technique, wherein trees are trained sequentially, with the newest tree correcting the errors of the previous tree. In both training techniques, each tree will come to a slightly different conclusion. The model produces a single, final result by summing the result that each tree in the model comes to. Figure 1 visualizes the way that a computer trains an XGBoost model.

In order to optimize decision tree models, the researcher must determine specific parameters, such as how many times the decision tree should be allowed to branch or how many decision trees the model should have. Too many trees results in the risk of overfitting the model, whereas too few trees will not fit the model closely enough to the data and leave a wide margin of error. Bayesian optimization techniques are used to determine the parameters that the model should use for optimal accuracy.

3 Related Works

A long history of data-driven traffic models that have been used to make planning decisions [2, 16, 10]. Data visualization methods plot a simple database of accident data onto a map to give the data an easily visualized, geospatial component. Other visualization models utilize BI graphics and tables to display accident causes or traffic metrics. These approaches cleanly visualize historical accident data but do not model or predict future accidents.

The traffic management field has historically relied on statistical regression analysis for decision-making, such as historical averages or ARIMA. Historical averages take the average

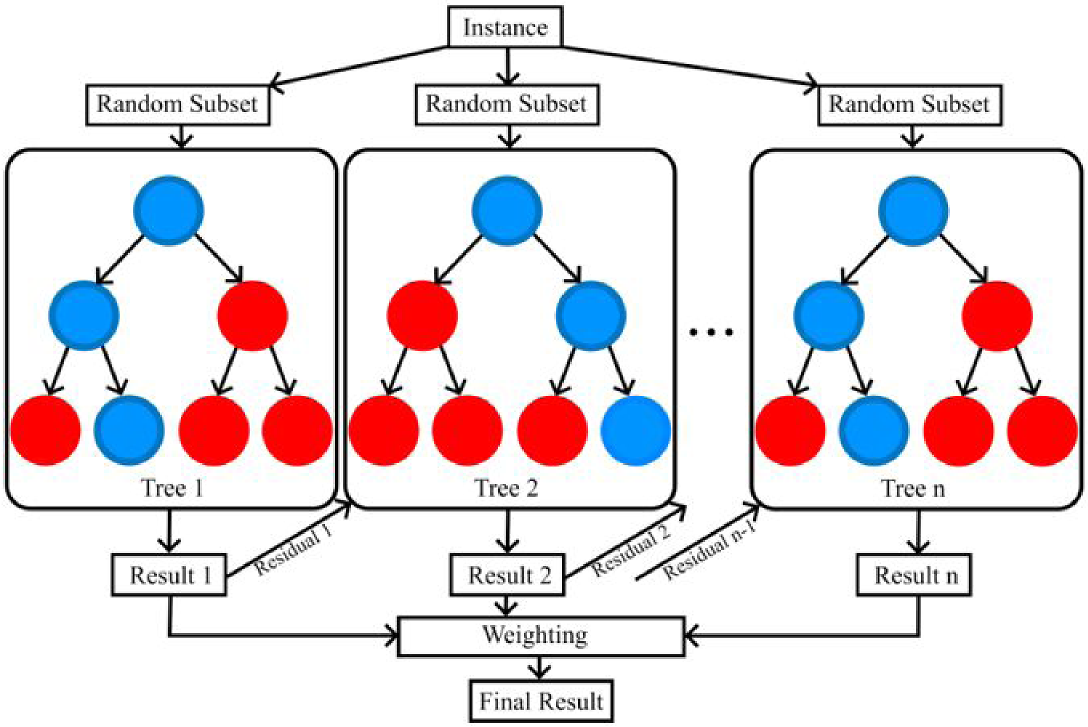

Figure 1: The XGBoost training method, in which a series of trees are built on random subsets of the data, each one learning from the previous one's mistakes.

of all data over a certain time period and use that value as a prediction of what will happen in the future. ARIMA, or Autoregressive Integrated Moving Average, provides a more complex analysis by integrating three components: "Autoregressive," which computes the current volume of traffic based on previous traffic counts (ex. if traffic is high at 10 am, it will likely still be high at 10:10 am), "Integrated," which handles broad trends over long periods of time, and "Moving Average," which allows the model to correct itself based on its past errors. These methods are computationally simple, but their simplicity limits their ability to handle non-linear relationships or the incorporation of additional variables, such as weather [11]. ARIMA, specifically, handles short-term time-based predictions well, but struggles with real-time traffic modeling, managing the impact of traffic flows from multiple locations, or handling data with a large variety of variables.

Recent approaches using machine learning to model traffic flows, jams, and accidents have shown a marked increase in accuracy of traffic modeling compared to traditional statistical methods [11]. Machine learning approaches are less computationally expensive than neural networks and excel at non real-time traffic analysis and prediction. A plurality of studies have observed the superior performance of machine learning models, especially decision tree algorithms, over neural networks such as LSTM or CNNs in the creation of accurate traffic models. Random Forest and XGBoost have similar accuracy, with some comparison studies finding an edge when using XGBoost [6, 1] and some finding Random Forest to be the most efficient [9, 15]. Decision tree models often do not create real-time traffic maps, but perform especially well on tabular data that is often available for modeling problems. K-Nearest Neighbors (KNN) and Support Vector Machine are other frequently used machine learning models used to analyze traffic with similar accuracy but more computation complexity. Machine learning traffic studies apply these methods towards a variety of goals: emission tracking [4], traffic jam forecasts [9], and accident prediction [2], but they share similar methods, machine learning techniques, and limitations in the creation of their traffic models.

Machine learning models suffer from a lack of interpretability; the model simply reads out the result without speaking on the factors that went into that decision. For instance, an XGBoost model may say that there will not be an accident on Beall Avenue tomorrow, but it will not be able to say that it made that decision because the weather will be clear, or because accidents occur less often on Wednesdays, or because it is springtime. A lack of explainability in machine learning models is both uninformative and seen as unreliable by policymakers who don't understand how the result came to be. In order to create interpretable results, studies use local interpretable methods like SHapley's Additive exPlanation (SHAP), as well as global feature importance methods like Permutation Feature Importance (PFI) and Feature Importance Ranking Measures (FIRM) ABDULRASHID2025103969. Local feature importance conveys information about why the model made each specific decision, whereas global feature importance explains which features are most important across the entire dataset. However, the effectiveness of each interpretability method may vary based on the machine learning method used, as Abdulrashid [1] noted that each feature importance model gave different results when used for each different machine learning method.

Neural networks provide another type of traffic modeling framework which performs especially well for real-time traffic analysis, such as in determining travel time or in adaptive signal control [11]. Neural networks comprise the backbone of Long Short-Term Memory (LSTM) and Artificial Neural Network (ANN) are commonly used and useful due to their ability to deftly handle spatiotemporal data. A study in Ali Mendjeli, Algeria uses an LSTM model to determine accessibility of public transport in real-time [7]. Another study utilizes ANN to create an adaptive signal control system in AlKharj city [12]. Neural networks are more computationally expensive than simpler machine learning techniques, but can better handle data required for adaptive, real-time transit software.

Traffic modeling studies share limitations and concerns. In any smart city project, privacy becomes an issue as personal travel data is being used to create traffic flows. Data privacy concerns contribute to the already herculean difficulty in acquiring the necessary data to create traffic models. Even when data is available, the data may not accurately represent the actual traffic patterns in the city. For instance, many studies use data acquired from Uber or Waze to create neural network-based models, but acknowledge that training a model based on only one transportation platform may be biased since it would not represent the traffic of all individuals within the city [9]. Finally, data availibility varies between cities and regions, which results in more data-rich areas receiving more researching resources. The majority of traffic accident models focus on large cities where data is more readily available, rather than small cities where different factors may contribute to accident occurance and severity. Urbanized cities in North America have a larger abundance of data available to train machine learning models or neural networks, such as Uber data, sensors, cameras, and satellite data. Developing cities face greater challenges in acquiring the data required to model traffic flows [6], and research projects often require collaboration with local governments in order to place sensors or otherwise acquire traffic data that can be used for analysis.

4 Methodology

The goal of this project was to determine what circumstances would strongly impact the severity of a traffic accident. To carry out this project, an interpretable machine learning XGBoost model was trained to forecast and analyze traffic accident severity in Wayne County, Ohio. The model was made "interpretable" by conducting a feature analysis to determine what features were most indicative of accident severity. The model was evaluated based on accuracy, precision and recall and compared with logistical regression models and the statistical average.

4.1 Data Gathering

The data were gathered from the Ohio Department of Transportation, which collects and records all traffic accidents that have been reported to law enforcement. Only data from the past five years at the time of the experiment was available, so only accident reports from January 2021 to December 2025 were used to train the model. The five year time frame nevertheless allows for enough statistical data to be used without going too far into the past to where the data is no longer relevant. Figure 2 plots the accident data to their geographical location, color-coded by severity.

Potential biases could lie in the inclusion of accidents to the dataset only when recorded by law enforcement, excluding numerous minor accidents or accidents in rural areas further from police stations.

4.2 Data Cleaning

The dataset was divided into five categories of crash severity. Figure 3 shows how the data overwhelmingly falls into the low-severity category of "Property Damage Only," while only a very small number contain serious injuries or fatal datapoints. A number of adjustments were made to the model to account for the uneven distribution of data, as described in 4.3.

The project was coded using Python, which is especially adept at handling machine learning models such as XGBoost due to its multiplicity of specialized packages available for installation and use. Raw data was transformed into data suitable for machine learning analysis through rigorous cleaning, selection, and transformation. Extraneous features from the original dataset that have no influence on crash severity, such as the police officer who reported it and their badge number, were removed from the dataset. The python packages pandas and scikit-learn were then used to prune and transform the remaining features to make them machine-readable.

Feature data was divided into numerical and categorical data to be evaluated and transformed. All values had to be encoded into numerical values and adjusted to be easily read into the model. The scikit-learn standard numerical encoder for numerical data, which ensured that all the numerical values were viewed with equal importance. For instance, it

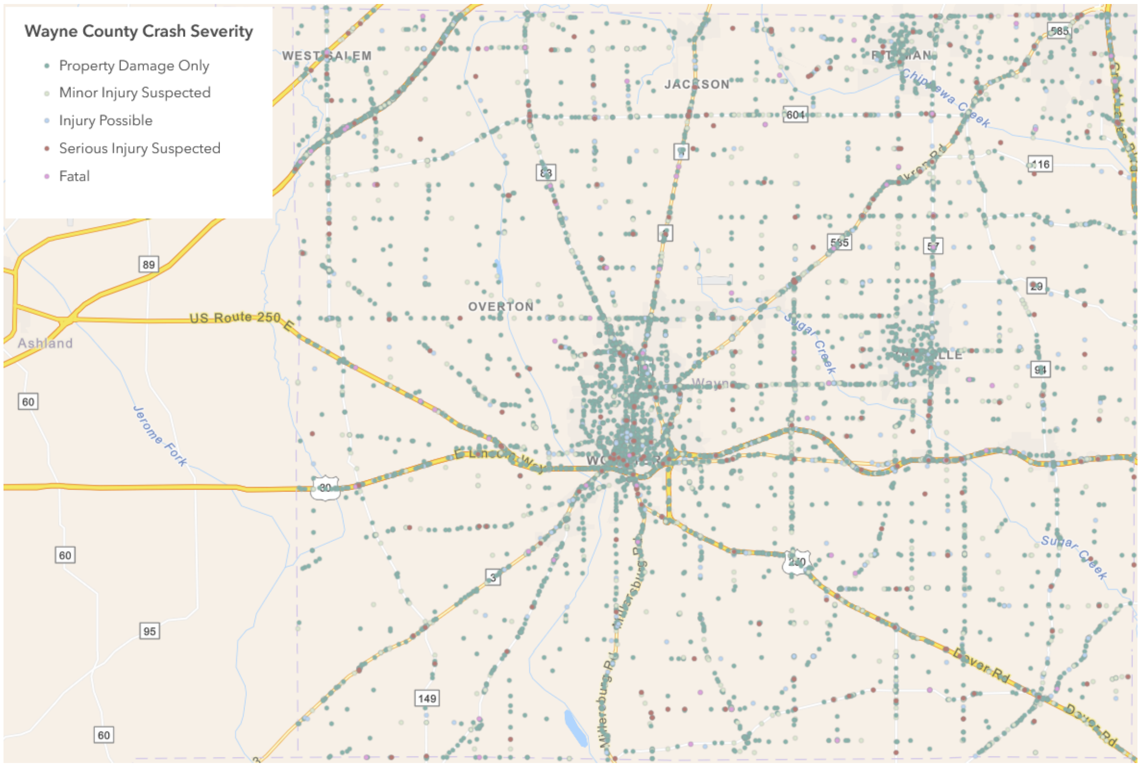

Figure 2: Accident data from 2021-2025 in Wayne County, Ohio.

ensured the model would not read features which store data in large numbers like 10000 as having more weight than features like "minute," which only go up to 60. OneHotEncoder was used for categorical data such as "LightCondition," which assigned a number to each category (such as "Cloudy" or "Rainy") so that the model can read and understand the categorical data. Missing data was ignored. For the "y" variable of crash severity, the scikit-learn label encoder performed a similar function of encoding categorical data to a number (0, 1, 2, or 3). The results of this encoding are displayed in Table

The scikit-learn class SelectFromModel pruned the remaining features to include only features with a logloss value 1.25*mean. Rather than including features with a logloss value above the median, the increased number of included features is intended to catch values that may be important in determining a less-represented, "fatal" crash category.

Category Name	Label
Property Damage Only	0
Injury Possible	1
Minor Injury Suspected	2
Serious Injury Suspected	3
Fatal	4

Table 1: A table descring the label encoding of crash severity.

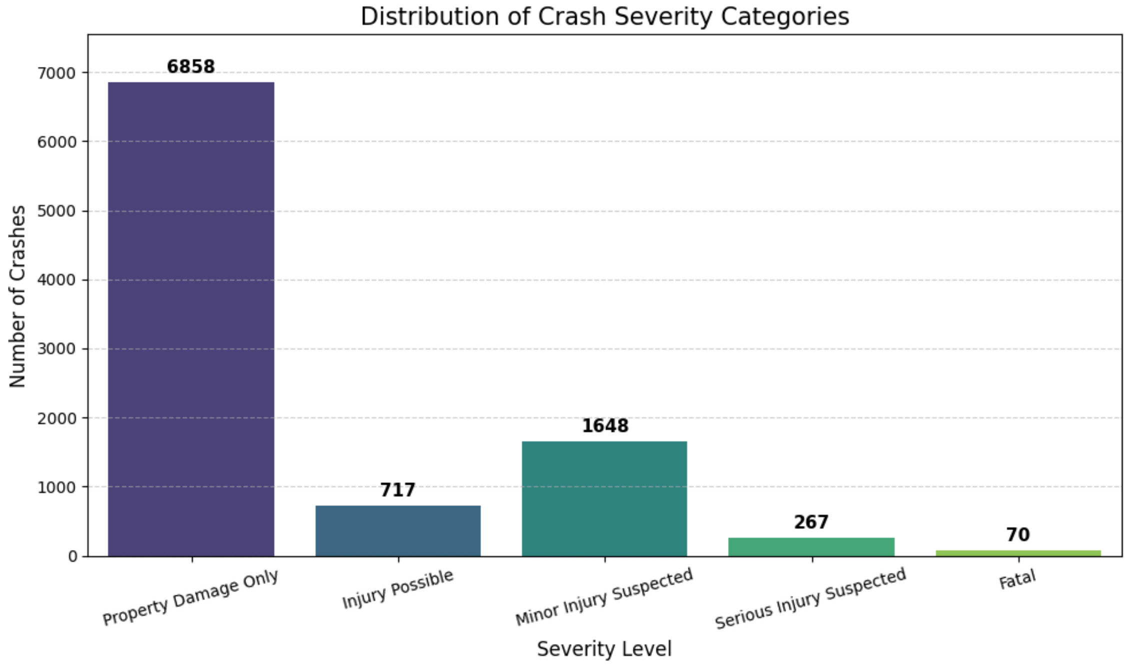

Figure 3: Distribution of datapoints for each crash severity category.

4.3 XGBoost Model

XGBoost was chosen due to its ability to analyze non-linear trends specifically within tabular data. In comparison, statistical average models like ARIMA are less adept at handeling non-linear relationships. Additionally, XGBoost is a relatively less complex algorithm that takes less time and computational power to train than a neural network like an LSTM. The built in feature importance component of XGBoost and its existence as a python package further lent the model to use for this project.

Training and testing data was split 80-20 and the XGBoost model was trained on the training data. To enhance abstraction in the code's design, a ML pipeline was used to combine data transformation and feature selection alongside the XGBoost model. This enabled the creation of a single pipeline which fit the fully processed, selected features to the model in one step.

The parameters for the XGBoost model were determined using a Grid Search. Grid Search is a hyperparameter optimization method used to tune the parameters of ML models– for example, the max depth of the trees, or the learning rate. Grid Search was chosen due to its simplicity and its ability to quickly find the best parameters when given a range of defined parameter values.

The model had to be adjusted significantly to account for the uneven data split inherent in the dataset. In a traffic accident dataset, the majority of the data had no or limited severity, while only a very small percentage contained serious injuries or fatalities, as shown in Figure 2. To account for the uneven data split, class weights were added when fitting the XGBoost pipeline to penalize mistakes in the rarer categories (those with more serious

Parameter	Options	Selected
'XGB__n_estimators'	[25, 50, 75],	75
'XGB__learning_rate'	[0.01, 0.05, 0.1],	0.1
'XGB__max_depth'	[3, 5, 7, 8],	5
'XGB__subsample'	[0.8, 1.0],	0.8
'XGB__colsample_bytree'	[0.8, 1.0],	0.8

Table 2: A table describing optimal parameters for XGBoost as determined by GridSearch.

outcomes) more heavily than mistakes in the more common categories. This ensured that the model focused more heavily on the factors that determined fatal outcomes to make up for the lack of representation in the dataset.

Since this was a classification problem, precision, recall, and f1 scores were used to determine the accuracy of the model. Precision measures the rate of false negatives, while Recall measures the proportion of true positives that were detected by the model. The f1-score represents the harmonic mean of the two values. Since these values are reported per crash severity category, they can be used to accurately evaluate the accuracy of the XGBoost model. This is especially valuable given the uneven data split of the dataset: these statistics report the accuracy of the XGBoost model in predicting each category rather than just the overall accuracy, which would be heavily biased towards the most common category.

5 Results

The output of the statistical baselines, the XGBoost model, and the visualization are displayed in the following section. The XGBoost model forecasted the severity of traffic accidents in Wooster, Ohio with a 35% accuracy rate.

5.1 Hyperparameter Optimization

The parameters optimized using Grid Search optimization methods are described in Table 2. A relatively low "n_estimators", number of trees used in the model, is determined at 75, with the learning rate of 0.1 showing that the model prefers a smaller number of trees that learn quickly. These trees are relatively short with a max depth of 5 nodes, subsampling 0.8 of the data each time. Additionally, the final test accuracy with the best model according to the grid search was 0.7202, at about 72%.

5.2 XGBoost

The classification metrics for the XGBoost model are compared to a "dummy" variable which sets every prediction to 0 (Property damage only), as shown in Figure 3. A table mapping the crash severity categories 0-4 to their respective names can be shown in Table 1. The comparison to a dummy variable is useful to determine how the XGBoost model's predictions compare to a base average value. Precision rating of 0.72 displayed for the most frequent "0" category indicates that the model at an accuracy of 72% as shown in section 4.1 before weighing each category evenly is no better than guessing the most common category for every data point

	precision	recall	f1-score	support
0	0.72	1.00	0.84	1372
1	0.00	0.00	0.00	143
2	0.00	0.00	0.00	330
3	0.00	0.00	0.00	53
4	0.00	0.00	0.00	14
accuracy	0.72	1912	0.35	1912
macro avg	0.14	0.20	0.17	1912
weighted avg	0.51	0.72	0.60	1912

Table 3: Classification metrics for "dummy" results can be used to compare the baseline model against an average value.

The table displaying the classification metrics for the XGBoost model are displayed in Table 4. These results represent the model trained with the XGBoost pipeline, including the data transformation, feature selection, XGBoost model with optimized parameters, and class weights. The accuracy rating displayed in Table 4 differs from the 72% described in section 5.1 to 35% when class weights are applied to the XGBoost model, and thus take into account the uneven distribution of data.

The most common "0" classification has a high precision score and a relatively low recall score, indicating that the model is identifying other crash severity categories when it should be identifying "0". In contrast, categories 2 and 4 show relatively low precision and high recall at around 52%. These measurements indicate that the model has overcorrected from its 72% accuracy when it guessed everything to be 0, and now triggers too often for categories 2 and 4. The low precision and recall scores for both 1 and 3 similarly indicate a systematic flaw in the model's decision making process.

5.3 Explanatory features

Figure 4 displays a graph of the features that mattered the most to the XGBoost model when the model was deciding how to categorize new data points. The global feature importance analysis shows that the DrinkingRelated = False category is by far the most indicative of crash severity. Additionally, other "Danger" boolean variables such as the presence of motorcycles, pedestrians, and animals/deer are also used by the model to predict crash severity. Various MannerOfCollision categorical variables also appear in relatively high importance.

Given the acute difference between the importance of the DrinkingRelated feature and the other features in crash severity prediction, it was pertinent to determine what outcome(s) the model determined based on the DrinkingRelated feature being marked false. Figure 5 visualizes the number of non-drinking-related-accidents that belong to each severity category, showing that the vast majority of non-drinking related accidents fall into the "Property Damage Only" category.

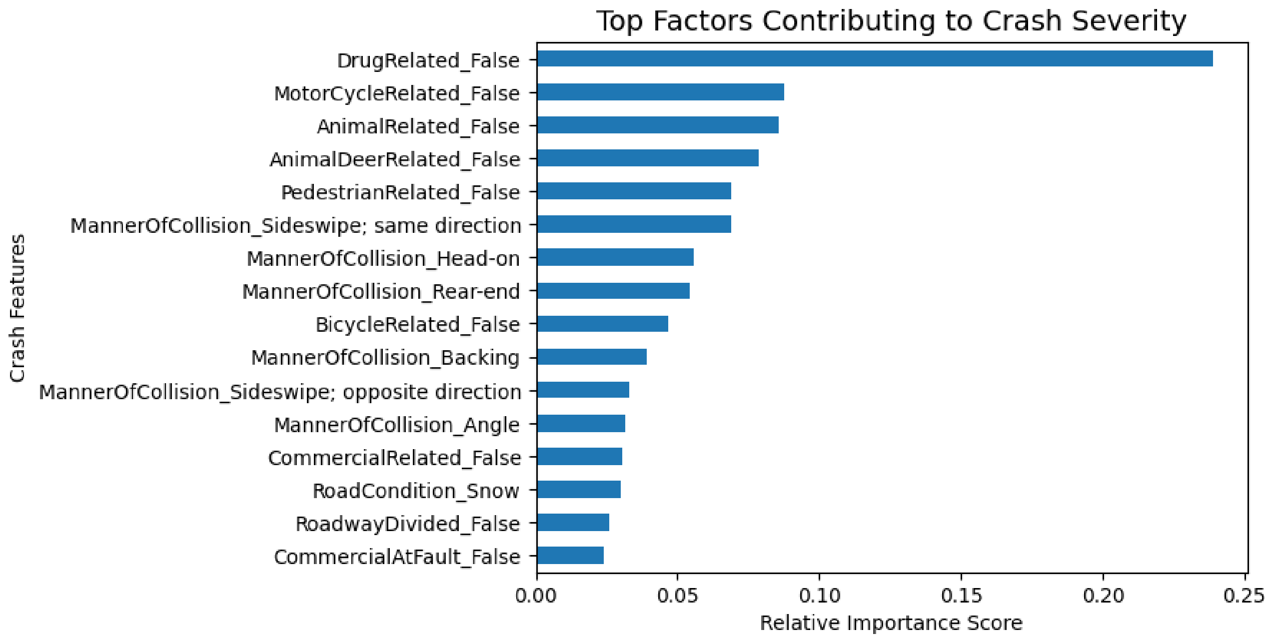

Figure 4: As displayed by the figure, the DrinkingRelated = False condition is the most indicative of accident severity.

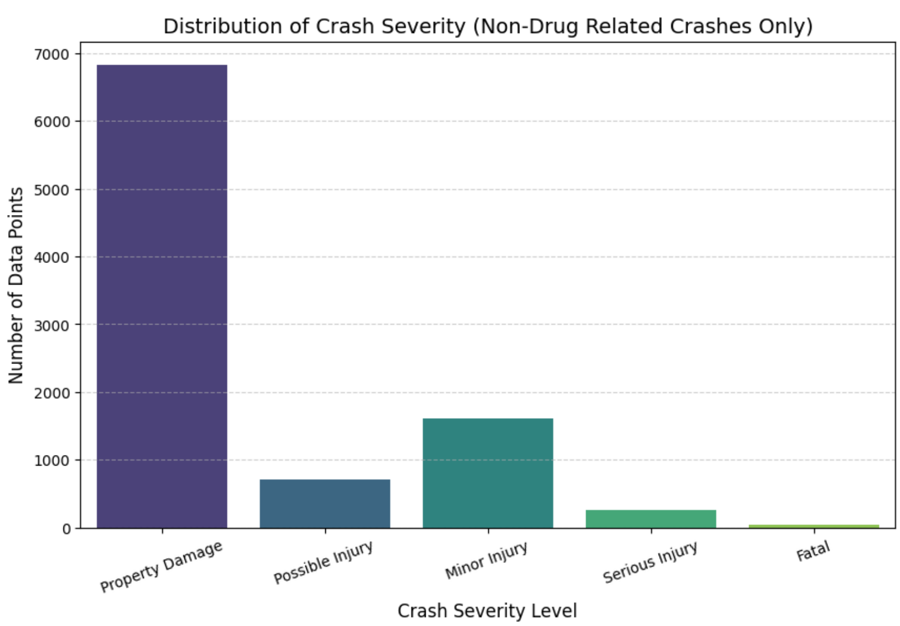

Figure 5: The DrinkingRelated = False condition is heavily correlated with less severe outcomes.

	precision	recall	f1-score	support
0	0.88	0.31	0.46	1372
1	0.10	0.31	0.15	143
2	0.20	0.52	0.29	330
3	0.19	0.28	0.23	53
4	0.13	0.50	0.21	14
accuracy			0.35	1912
macro avg	0.30	0.39	0.27	1912
weighted avg	0.68	0.35	0.40	1912

Table 4: The table displays precision, recall, and f1-score of the XGBoost model. The "support" column contains the total number of datapoints for that category. "macro avg" takes an average each category equally regardless of its "support", while "weighted avg" calculates the average with respect to its "support".

6 Analysis and Discussion

The project's implmenetation of the XGBoost model resulted in a low accuracy rating and inconclusive results. The following sections compare the results found in the project to similar projects in the literature to determine the outcomes.

The classification metrics showed how the XGBoost model generated inconclusive results, primarily because it was evaluating based on an uneven spread of data. Other similar studies [2] which did traffic analysis made sure to have a dataset with an even distribution of datapoints for each category so as to not face a similar problem. A greater pool of data over a wider range would be needed to generate enough datapoints to evenly have a distribution of fatal, injury, and property damage categories. This study also improved their metrics by using different categories of crash severity, usually a three-fold category: No injury, injury, and fatal. A similar scheme can be implemented to improve accuracy of the model.

Our results found that non-drug related accidents were by far the most useful indicator of a non-serious traffic accident. These results are supported by general crash statistics, which state that about 30% of all traffic fatalities are caused by drug-related accidents [13]. Previous statistical analysis have also found that certain types of collisions, such as head-on collisions, are more likely to result in severe accidents.

7 Conclusion

The project sought to determine the factors that contribute most to traffic accident severity. From a development perspective, it sought better ways to train ML models and ways to visualize ML traffic models in a way that is useful to policymakers. The research contributed to a growing body of literature which uses machine learning techniques to analyze traffic accident data. It found that non-drug-related-accidents were by far the most likely to have less severe traffic accident outcomes. MannerOfCollision was also a key determinant of traffic outcomes. These results are supported by general crash statistics, which state that and that certain types of collisions, such as head-on collisions, are more likely to result in severe accidents. These results can be generalized to assist policymakers and urban planners in ways to decrease the severity of traffic accidents.

The project was limited by a lack of data, especially data of higher accident severity. The low accuracy rating of the model indicates the possibility for inaccurate results predicted by the model and a potential threat to validity. Further, the scope of the data was limited by time constraints and by the inability to obtain traffic count data that could have been used for traffic modeling and limited the analysis to accident severity prediction.

The model had a low accuracy rating at 35%. Future work to improve this model can make small adjustments to the model, such as cyclical encoding of numerical variables, different handling of null values, or different categorical groupings than what was initially described in the data. The work may also be inhanced by the addition of data, especially data in underrepresented crash groupings (like fatalities) so that each category is evenly represented. This can be done by increasing the geographical or temporal sample of the data. If XGBoost continues to produce low accuracy results, a change to a different model such as Random Forest of K-Nearest Neighbors may prove more fruitful.

Future research in traffic analysis can expand on this research through different types of analyses, such as geographical or other statistical analyses. Additional geographical/traffic count data can be used in conjunction with this model to predict accidents based on real-time traffic flow or detect traffic accident hotspots for further analysis. SHAP analysis can be applied to create a more robust feature importance analysis which can accurately state which features most contribute to the model choosing one category or the other, rather than the built-in global feature importance ranking from XGBoost.

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